

Why AI Guidelines Matter...

A national initiative of the Department of Science and
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...too prevent this...

...AI From Taking Our Jobs...

...As it is assumed they are better than humans...until those same humans then get rehired...

Ford brings back 300+ **human engineers after AI fails** to match their skills & experience

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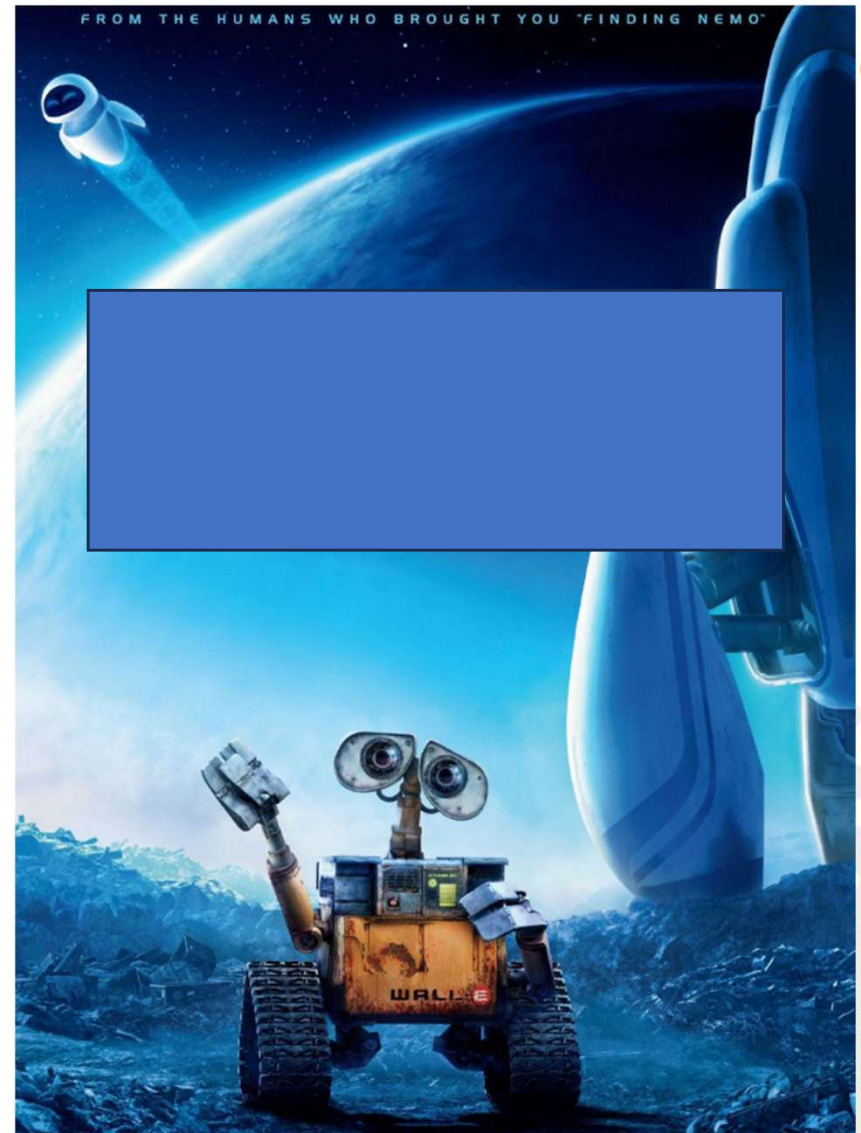


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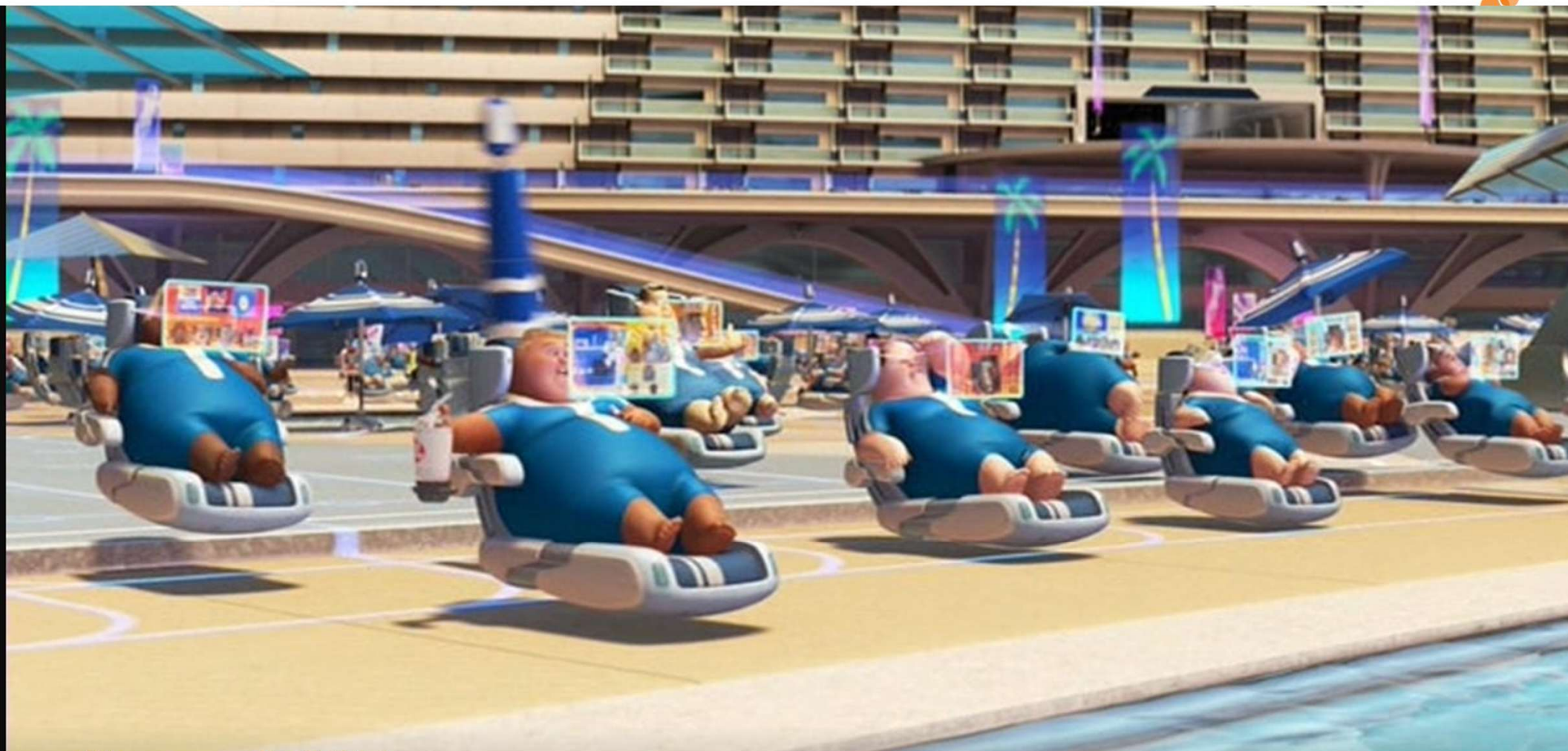


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...and to prevent this...







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Why AI Guidelines Matter



- AI assistants and coding copilots are everywhere
- AI now touches infrastructure, not just code
- This introduces new operational and security risks
- Governance must become practical, not theoretical

“Insurance company chatbot... ☹️”

3 Levels of AI Autonomy

- Level 1 — Human-in-the-Loop (HITL)
- Level 2 — Human-on-the-Loop (HOTL)
- Level 3 — Human-out-of-the-Loop (HOOTL)

Level 1 — Human-in-the-Loop



Human makes final decisions and executes actions.

- AI-assisted coding
- Documentation generation
- Debugging assistance
- Chatbots, ChatGPT, Copilot etc...
- These slides 😊

“Lowest risk. Human remains
in control.”

Level 2 — Human-On-The-Loop

AI performs actions autonomously while humans monitor and intervene when needed.

- AI CI/CD assistants
- Automated cloud remediation
- AI infrastructure monitoring
- Cursor Agent, Claude Code etc...

“Moderate risk. Requires strong oversight.”

Level 3 — Human-Out-Of-the-Loop



AI operates independently with little or no human involvement.

- Fully autonomous agents
- Self-healing infrastructure
- AI-to-AI orchestration
- OpenHands, Open WebUI, ...

“Highest risk. Dangerous in critical infrastructure.”

3 Levels of AI Autonomy

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The Shift

From:

- AI as a chatbot

To:

- AI writing code
- AI running scripts
- AI accessing APIs
- AI controlling infrastructure

*“YouTube Add – Don’t use AI
like you google...but I like to
code !”*

What is IaaS?

Infrastructure-as-a-Service includes (IaaS):

- Cloud virtual machines
- Storage systems
- Networks
- HPC clusters
- CI/CD systems
- Research platforms

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Two Sides of AI Governance



Shared Responsibility:

1. AI Users
2. AI Providers

Both need guardrails.

*“Policies, governance,
guidelines...sounds pretty
boring!”*

AI Users

Examples:

- Developers
- Sysadmins
- Researchers
- DevOps engineers

They use AI to:

- Write code
- Debug systems
- Automate tasks
- Query infrastructure

Risks from Users

Common risks:

- Credential leakage
- Blind trust in AI output (Biases)
- Unsafe scripts
- Wrong commands
- Over-permissioned agents
- ...ability to think...

“...Can you replace AI”

AI Providers

Examples:

- Cloud providers
- HPC centres
- Platform teams
- Infrastructure operators

“They provide the environments AI can access.”

Risks from Provider Side

Examples:

- Excessive AI permissions
- Missing audit trails
- No sandboxing
- Weak policy controls
- Poor monitoring

“They provide the environments AI can access.”

Real Example

Prompt: “Fix my cluster issue”:

AI may:

- Kill jobs
- Delete files
- Modify configs
- Restart services
- Expose OS, Network vulnerabilities etc...

Fast ≠ Safe

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South African AI Policy Incident



In 2026, South Africa withdrew its draft national AI policy after fictitious AI-generated citations were discovered.

NEWS

SOUTH AFRICA

Minister Solly Malatsi's withdrawal of draft national AI policy for its fabricated sources 'embarrassing and damning'



[Theolin Tembo](#) | Published 2 weeks ago



4min

South African AI Policy Incident



Key lessons:

- Which level of AI was used?
- AI output is not verified truth
- Human review is mandatory
- Governance applies even to policymakers
- AI can accelerate mistakes at scale

*“If AI can hallucinate inside AI policy itself,
what happens when AI touches
infrastructure?”*

From Theory to Practice

What Are We Actually Doing?

At the CHPC, AI governance is approached in layers:

1. Organization-wide AI policy
2. Infrastructure-specific AI policy
3. Technical enforcement controls

“Policy alone is not enough. Controls must enforce policy”

CSIR AI Policy Foundation



Broad AI governance principles include:

- Responsible and ethical AI use
- Human accountability remains mandatory
- Protection of sensitive data
- Legal compliance
- Transparency in AI-assisted work

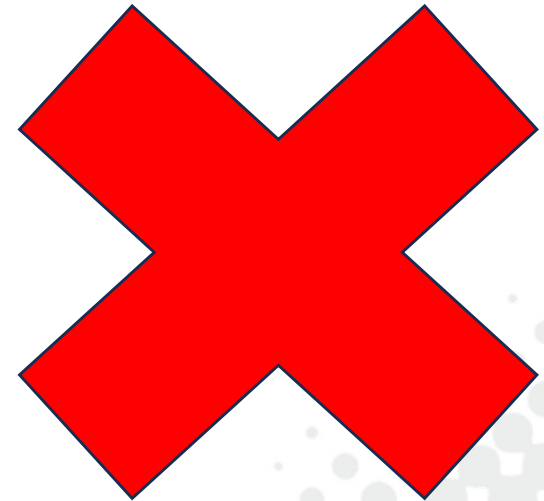
“This provides the high-level governance framework.”

Why Generic AI Policy Is Not Enough

Infrastructure teams need more specific rules:

- Can AI access SSH credentials?
- Can AI trigger deployments?
- Can AI access production databases?
- Can AI execute privileged commands?

Infrastructure needs explicit guardrails.



Data Sovereignty Matters

AI governance is not only about access control.

It is also about:

- Where data lives
- Where computation happens
- Which legal jurisdiction applies
- Who controls the infrastructure

“Key question...Where is your AI processing your data?”

Convenient ≠ Safe

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Data Sovereignty Matters



At CHPC, governance is not just theoretical.

We apply it through infrastructure:

- HPC cluster (Lengau)
- Local cloud (Sebowa)
- South African data centres (Cape Town and Pretoria)
- As a researcher, this service is for FREE!

This allows users to run workloads without exporting sensitive data offshore.

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Local Infrastructure

With local HPC or cloud infrastructure, we can support:

- Data remaining in South Africa
- Better control over storage and access
- Compliance with local governance requirements
- Reduced exposure to foreign legal frameworks...geopolitics

Instead of sending data to a random foreign platform, we build infrastructure where controls are enforceable and you are in control!

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CHPC AI Policy

Infrastructure-focused AI policy includes direct operational rules:

- No AI agent/tool may store or access private SSH keys
- No AI agent/tool may receive root or sudo credentials
- No autonomous production changes without approval
- No AI agent/tool may access sensitive user data by default
- All AI actions must be logged and auditable

These are easier to enforce for staff but more challenging for users

CHPC AI Policy



These controls can be directly enforced for infrastructure staff through technical and administrative controls.

For end users, complete enforcement is not possible!

Therefore, the infrastructure rules applies the following to everyone:

- Least-privilege access
- Network Isolation
- Monitoring and auditing
- Promoting secure AI usage through awareness

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Practical Application: Try Local Infrastructure!



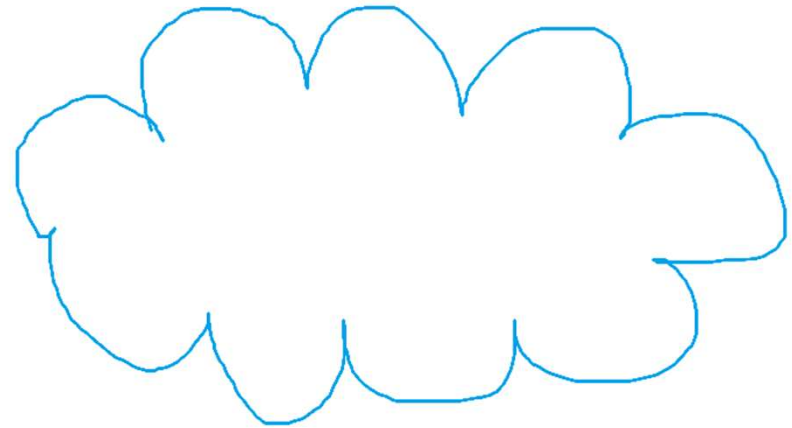
A practical alternative is to run VMs, databases, or AI workloads **on local South African infrastructure.**

Through CHPC cloud infrastructure you have multiple options:

1. HPC Cluster Lengau – CPU and GPU Compute
2. Sebowa Cloud – Openstack Self Service Cloud Portal
3. VuthaStack – Spin up VM's and pre-built stacks with in minutes

Sebowa Cloud

- Self-service Openstack cloud portal to launch and manage VMs.
- Networking and floating IPs.
- Security groups and SSH keys.
- South African hosted infrastructure.



“Link at the end of presentation”

VuthaStack: VMs in minutes



Quickly evaluate services before fully comprehensive Sebowa Openstack environment

Examples:

- VM Only: Ubuntu, Rocky Linux, CentOS VM
- GPU-enabled (NVIDIA A100) VM, and development sandbox
- Pre-built stacks: Chatbot, JupyterLab, Kubernetes

This gives users hands-on access to South African infrastructure in a controlled environment.

Closing

Responsible AI infrastructure requires three pillars:

- Governance: what AI is allowed to do
- Security: what AI is allowed to access
- Sovereignty: where AI is allowed to run

Through supporting local infrastructure, AI governance becomes practical - not just policy.

In infrastructure, “trust” must be earned — never assumed.



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Thank You!

*Developing AI Guidelines for
Infrastructure-as-a-Service
Platforms – IndabaX 2026*

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